INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A** Semester: **Third**

Stream: **BCA**

Paper Code: **BCAAE301**

Paper Name: **Environmental Studies**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1.	Mention two importance of atmosphere.	**2**
2.	Report the chemical compounds present in acid rain.	**2**
3.	Show the distribution of water on the earth with diagram.	**2**
4.	How convection currents in earth mantle created?	**2**
5.	Explain the reason behind the seeding of BOD test.	**2**
6.	Represent Darcy's law.	**2**
7.	Name the four R's principle in solid waste management.	**2**
8.	Name the pollutants related to itai-itai and minamata disease.	**2**
9.	Give examples of four ecosystem components.	**2**
10.	Differentiate between closed system and open system.	**2**
11.	Represent sound intensity in decibel.	**2**
12.	What is the full form of dBA?	**2**
13.	What is speciation of metals? give one example.	**2**

14. State the characteristics of hazardous wastes. **2**

15. What are toxic metals? Give two examples. **2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Write brief notes on dissolved oxygen content in aquatic systems. **5**

17. Represent the characteristics of lithosphere. **5**

18. Describe the processes of formation of acids in atmosphere. **5**

19. 50 ml of waste water is filled in 300 ml bottle with unseeded diluted water. This solution shows a drop of DO to 5 ppm after 5 days. What should be the BODs of the water? **5**

20. Explain the mechanisms of rotating biological Contactor (RBC) in waste water treatment. **5**

21. Represent types of soil pollutants and transport processes of pollutants to soil. **5**

22. Discuss on the characteristics of pyramids of biomass in pond ecosystem. **5**

23. Outline various types of decomposition of organic detritus in nature. **5**

24. How can noise pollution be controlled at the source and along its path? **5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Describe the processes of controlling the particulate matter emissions in the atmosphere. **10**

26. Outline the merits and demerits of the presence of particulate matter in the atmosphere. **10**

27. Write a note on ground water aquifer covering with types, mode of formation, ground water level, soil porosity and hydraulic gradient. **10**

28. Describe trickling filter mentioning advantages and disadvantages along with diagram. **10**

29. Describe nitrogen cycling explaining different steps with diagram. **10**

30. Discuss on oxygen cycling occurring between different environmental segments. **10**

31. Discuss on the characteristics of transport noise and occupational noise in the environment. **10**

32. Outline general characteristics and various units of sound together with indexes used to represent noise pollution. **10**

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